Sustainability of a university's quality system: adaptation of the EFQM excellence model

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Received 10 September 2019
Revised 24 October 2019
Accepted 24 October 2019

Abstract

Purpose – Modern trends show that universities are searching for new solutions to increase efficiency and improve quality by considering approaches of quality system development that link with strategy and include extensive analysis of risks, processes and stakeholders. The approach that best fits the institution has to be in line with the institution's strategic objectives, quality culture and policy and key performance indicators. The purpose of this paper and case study is to discover if sustainable development may be achieved by using an appropriate quality system development approach, such as the European Foundation for Quality Management (EFQM) excellence model.

Design/methodology/approach – To analyse sustainability and development of the quality system adapted in a higher-education institution, a literature review of different quality management approaches and models was performed. The research includes a case study of the university's quality management framework based on an adapted EFQM excellence model emphasising on strategic development in the context of sustainability. The key focus of the research is to discover how universities could better focus on sustainable development and benefit from a quality system based on an adapted EFQM excellence model.

Findings – Literature analysis indicated that some of the sustainability development activities a university may use are possible to be integrated through quality system models and development approaches. The findings from the literature suggest that the EFQM excellence model may provide a management framework and a comprehensive overview of a university for identifying necessary improvements and promoting the implementation of advancement activities on the road to sustainable development. The principles of the EFQM excellence model may guide in setting a strategic focus on sustainable development of universities.

Originality/value – Sustainability and sustainable development of a university are analysed in terms of quality aspects of higher education, and the research results reveal the main sustainability elements to be analysed and implemented through a university's strategy and improvement processes. The Riga Technical University quality system called "RTU Excellence Approach" development analysis is given as a case study.

Keywords Sustainability, Quality, Quality management practices, Strategic management, EFQM model, Excellence model

Paper type Case study

This paper forms part of a special section "QMOD-ICQSS Conference".

The authors would like to thank Professor Su Mi Dahlgaard-Park and Professor Jens J. Dahlgaard for valuable guidelines in completing this study.

International Journal of Quality and Service Sciences
Vol. 12 No. 1, 2020
pp. 29-43

1756-669X
DOI 10.1108/IJQSS-09-2019-0108

Introduction

Various sustainability-related activities can be an important part of a university's operations. However, to manage them more effectively, these activities should be integrated into a system. Modern trends show that universities are looking for new systematic solutions to ensure efficient and sustainable management. Preferences are given to quality system models and approaches that are best integrated and in line with the university's strategic goals, key performance indicators (KPI) and quality policy, and the search for innovative approaches has led universities to adapt business excellence models and practices. Research shows that sustainability-oriented innovative practices positively influence the performance of an organisation (Maletič *et al.*, 2014; Frolova and Lapina, 2015; Zeps and Ribickis, 2016), and organisations that adapt such new approaches are more likely to improve their management system and institutional performance (Esteban-Lloret *et al.*, 2014; Berman *et al.*, 1999).

The foundation of sustainable process management is developing a strategy for organisational sustainability (Perrott, 2015) and, hence, sustainable and integrated process management as a concept is an opportunity to manage the university's resources more effectively in a competitive environment (Gough and Scott, 2007). Sustainable development requires the need for more accurate measurements of the university's physical, environmental and human resources to ensure fact-based decision-making (Rocha-Lona *et al.*, 2015), and systematic and effective management of processes is closely linked to the overall university performance achievements and sustainability (Balzer, 2010).

Literature review indicates that quality system models based on an appropriate business excellence model, such as the European Foundation for Quality Management (EFQM) excellence model, may be used to identify, measure and manage sustainability indicators of an organisation (Nemetz, 2014). Research has proved that it is possible to achieve positive sustainability development by implementing the EFQM excellence model in a broad variety of organisations (Rocha-Lona *et al.*, 2015; Ahmed *et al.*, 2003; Aryanasl *et al.*, 2016).

The aim of this research is to discover if sustainable development may be achieved in higher education by using an appropriate quality system development approach such as the EFQM excellence model. The key focus of the research is to discover how universities could better focus on sustainable development and benefit from a quality system based on an adapted EFQM excellence model.

Sustainability and development of quality system in higher education context

Nowadays different types of higher education institutions have been mentioned as key players ensuring the next generations of graduates to be more sustainability oriented. Sustainable development is not only what a university could invest into society and environment; it is also an important goal for the university's management itself. Being sustainable means managing processes and resources effectively (Rocha-Lona *et al.*, 2015), and to do so, sustainability and responsibility must be one of the values of the organisation (Isaksson, 2014).

While the United Nations World Commission on Environment and Development report (United Nations, 1987) gives a holistic definition of sustainable development, it does not discuss the necessary actions an organisation should do to become sustainable. The success of sustainability-related activities depends on higher management support and the ability of personnel to understand the need for change (Tuominen, 2011a, 2011b). For the success of implementing changes, it is important to develop a work culture that supports the principles and practices behind the changes (Snyder *et al.*, 2017). Changes may be challenging and therefore personnel involvement in decision-making may reduce resistance and improve adaption to changes (Iljins *et al.*, 2015). Leadership and maturity of the university's quality culture play a major role in the adaption process (Zink, 2007; Marshall *et al.*, 2017; Lapiņa *et al.*, 2015). Integrating sustainability concepts in the core of the university's strategy allows combining corporate social responsibility (CSR) indicators with the main strategic objectives (Tuominen, 2011a, 2011b). Also, benchmarking with other universities and even organisations of different spheres is an important first step to ensure sustainability (Dupada *et al.*, 2013).

The literature review indicated some of the activities a university may consider for organisational sustainability:

- identify *quality and sustainability leaders* at all university levels (Marshall *et al.*, 2017);
- ensure *higher management support* (Tuominen, 2011a, 2011b; Ozoliņš *et al.*, 2018);
- ensure *personnel awareness* about sustainability activities (Tuominen, 2011a, 2011b);*focus on stakeholders* needs (Zink, 2007; Lapiņa *et al.*, 2015; Marshall *et al.*, 2017; Degtjarjova *et al.*, 2018);
- ensure comprehension of the main *sustainability elements* (Tuominen, 2011a, 2011b; Ņikitina and Lapiņa, 2017);
- *integrate sustainability* in the university's strategy (Mežinska *et al.*, 2015; Zeps and Ribickis, 2016; Cazeri *et al.*, 2018);
- develop a CSR strategy (Cazeri *et al.*, 2018);
- identify and measure *sustainability KPIs* (Radoslav and Jankalova, 2016; Medne and Lapiņa, 2019);
- analyse *KPIs results* and compile an improvement plan (Tuominen, 2011a, 2011b; Iljins *et al.*, 2017); and
- *benchmark sustainability* best practices (Dupada *et al.*, 2013).

Universities may have different understandings of the above potential sustainability activities, but a balance of key sustainability elements – environment, economy and society – must remain. As universities' existing quality systems always are a manifest part of the university's operational culture – good or bad – it is important to ensure sustainability of the system, which only can be achieved by choosing capable and good leaders and by using appropriate methods and approaches for systematic management and improvement of key processes.

These are just a few of many potential activities to be addressed aiming to improve the organisational sustainability of a university. Furthermore, it is important not only to carry out these activities once but also to integrate them as continuous action plans into the university's processes and management. For this reason, it is necessary to choose an appropriate quality system model or development approach that integrates the key areas of sustainability and is in line with the university's strategic direction.

We, in our literature review, have analysed the most appropriate quality system models/ approaches adapted in higher education, and Table I below shows the analysed models/ approaches in column 1, the university context in column 2, and author names including publication year in column 3.

Total quality management (TQM) explains the holistic concept of quality management by defining and focussing on the basic quality principles. In the ISO 9000 standards, quality is defined as a relative concept, which is analysed through linking quality to requirements (Dahlgaard and Dahlgaard-Park, 2015). The use of the ISO 9000

Model/approach	Higher education context	Author, year
TQM	Holistic management approach including all management levels of the university. Focus on leadership, staff involvement and the benefits of society	Sangeeta *et al.* (2004)
Deming's 14 points	14 transformation principles that focus on continuous development using the PDCA/PDSA cycle. Focus on 1. Reduction of variation in quality and 2. Improving productivity and competitiveness	Ballard (2013), Dahlgaard (2015)
EFQM excellence model	Business excellence model to evaluate university activities by using nine criteria (five enabler criteria, four results criteria) and self-assessment, to determine progress of the university on the path to excellence	Bou-Llusar *et al.* (2009)
The Malcolm Baldrige National Quality Award	Business excellence model to evaluate university activities using seven criteria and self-assessment to determine progress of the university on the path to excellence	Bou-Llusar *et al.* (2009)
ISO 9000 standard	Quality is determined as compliance with a set of standard requirements. Focus on processes	Bouayad (2013), Dahlgaard and Dahlgaard-Park (2015)
Integrated management system (IMS)	System integration into the university's management system – quality, environment and safety. Focus on integration	Blanco-Ramirez and Berger (2014)
Balanced scorecard	Analysis of four university measurement perspectives – financial, student, internal processes and growth opportunities. Focus on result analysis	Hladchenko (2015)
SERVQUAL	Analysing stakeholder perceptions and expectations. The gap between the expected and received service. Focus on meeting and exceeding stakeholder needs	Vauterin *et al.* (2011)

Table I.
Quality system models and development approaches in the context of higher education

management standards and the development of an integrated management system (Mežinska *et al.*, 2015) are both based on compliance with quality requirements specified in the standards.

Deming's 14 points provide basic principles for transformation of industry (Tambi, 1999). Deming's plan-do-check-act/plan-do-study-act (PDCA/PDSA) cycle has been widely referred to and used in designing quality systems as well as business excellence models (Ballard, 2013; ISO, 2015; Bou-Llusar *et al.*, 2009). Examples of successful implementation of TQM in industry should be used as inspiration for other organisations (Dahlgaard-Park, 2015). Some universities have been able to adapt and integrate business excellence models such as the EFQM excellence model and Malcolm Baldridge Award Model into their quality systems (Ballard, 2013; Niedermeier, 2017).

The most popular method/approach competing with the mentioned excellence models is the Balanced Scorecard, which in a university context may be combined with the SERVQUAL model (Hladchenko, 2015; Vauterin *et al.*, 2011).

Some authors see obstacles for business model integration in the higher education context (Rocha-Lona *et al.*, 2015; Anastasiadou and Poulcheria, 2015). The main concerns are about the critical success factors, such as stakeholder and client definition, sufficient model understanding and correct determination of KPIs (Tuominen, 2011b).

The research shows also that some universities in Europe include references to EU policies in their strategies (Ozoliņš *et al.*, 2018). Some references can also be found in the

EFQM excellence model, for example, developing cooperation, managing university governance, quality assurance and introducing innovations.

Excellent educational institutions assess their operations regularly by using a consistent approach in line with the university's strategy and main objectives (Tuominen, 2011a, 2011b). By doing so, the university may be able to predict aspirations and demands generated within their operating environment. Trends show that universities are now searching for new solutions and approaches for increasing efficiency and improving quality by considering appropriate quality models with extensive assessment capabilities of their processes, stakeholders, risks and much more, and they have been adapting a variety of new approaches and models to improve their quality systems.

The results of our analysis of the literature indicate that a quality management framework, such as the EFQM excellence model, gives new opportunities to develop the quality management system according to the changing needs of a university and its stakeholders. The EFQM excellence model emphasises regular self-assessment in nine areas of the university thus providing systematic change management toward the development of system and process management. However, it is also important to determine how the EFQM excellence model is able to contribute to the implementation of sustainable activities in a university at the strategic level.

Analysis of the EFQM excellence model and sustainability factors

As this paper is on the adaptation of the EFQM excellence model focussing on strategic development in the context of sustainability of a university's quality system, we analyse the linkage between the EFQM excellence model criteria and university sustainability factors.

Before analysing the EFQM excellence model, it is important to understand how universities could achieve organisational excellence. The EFQM excellence model does not give a recommended set of suitable methods and approaches for achieving organisational excellence. Instead, the model focuses on determining the specifics of the university and invites to identify and implement the most suitable methods and approaches. The model helps to understand, through self-assessment, whether the chosen approaches will fit the university's context and will be efficient.

A quality management system provides a framework for sustainability element implementation in all management levels of an organisation (Frolova and Lapina, 2015). However, sustainability of a quality system cannot be achieved just by defining and implementing various approaches that seem feasible on paper. Such approaches should be regularly assessed and if necessary improved/changed. The right set of indicators and measures allows to track changes, errors and growth. A continuous development and learning cycle ensures planned activities to improve such approaches. In a rapidly changing environment, sufficient data has a crucial importance to building a sustainable quality system.

Some authors (Ballard, 2013; Rocha-Lona *et al.*, 2015; Anastasiadou and Poulcheria, 2015; Iljins *et al.*, 2017; Niedermeier, 2017) analysed the impact of implementing the EFQM excellence model in higher education institutions, and they identified various sustainability elements of the EFQM excellence model.

The EFQM excellence model defines nine criteria divided into *Enablers* and *Results*. The enabler criteria have the greatest impact on the development of the quality system (EFQM, 2013). This is why we, in our analysis, focus only on the first five criteria: leadership; people; strategy; partnerships and resources; processes, products and services.

The above authors have determined core sustainability elements in each of the five criteria of the EFQM excellence model. These elements complement the university's quality system on the road to sustainability (Table II).

The first analysed *Enabler* of the EFQM excellence model is *Leadership*, which has five important elements related to sustainability. Integration of sustainability elements in the core values and objectives of the university is a holistic approach to ensure sustainability at all organisational levels. To manage that, the university's higher management should understand and support sustainable activities. An important task is to understand and identify leaders as change agents at all organisational levels of the university. Leaders are employees who have a main role in different change projects throughout the university.

Planning activities for main stakeholder life cycles shows how well the university deals with internal and external customers, such as employees and students. There should be a complete understanding of what processes affect which stakeholder groups. In addition, such analysis should be taken into account when developing the university's strategy and activity plan. One of the main challenges is to address stakeholders' needs while developing the strategy. The foundation of sustainable strategy development is to understand the university's core processes and the needs of main stakeholder groups. University processes should be future-oriented and consider their effects on environmental aspects. If the university is developing a long-term strategy, then existing and potential stakeholders play an important role in ensuring sustainability.

From a quality system perspective, the university's strategy and activity plan should be viewed together with the system, process and human risk management. As organisational culture is a feature of a sustainable quality system, so should be all activities regarding personnel. Employees' active life cycle in the university starts with attracting potential employees and recruitment procedures. That is why a long-term personnel policy should be developed including analysis of current employees as well as potential/future employees. Analysing and developing processes related to personnel ensures the right conditions for employee satisfaction and employee growth. As a result of creating a sustainable organisational culture, the university succeeds in motivating employees to engage in decision-making and to undertake more responsibility.

One of the possibilities to benchmark best practices is to collaborate with partners that have proven to be excellent and sustainable. It is important not only to benchmark best practices but also to collaborate with partners. Developing sustainable and long lasting partnerships is essential for any university. Universities that collaborate and make suggestions to improve their partner institution processes are more likely to ensure sustainable partnerships. Processes

Table II. Sustainability elements in the enabler criteria of the EFQM excellence model

1. Leadership	2. Strategy	3. People	4. Partnerships and resources	5. Processes, products and services
Understanding sustainability Sustainable objectives and values Focus on stakeholders Life cycle Pilot projects for change	Integrated quality system Process integration Future stakeholder needs Consideration of environment	Personnel policy Engagement and responsibility Individual development plan Safety and risk management Organisational culture best practices	Sustainable partners Improve partners Requirements for suppliers Risk prevention Use of technologies	Process approach Process requirements Measurements System of needs Feedback

related to collaboration, for example, are student and employee mobility and internships with other universities or other employers. It is also important, if possible, to set requirements for suppliers or partners. The university may reduce a number of risks for lack of sustainability just by effectively managing partnerships and resources.

Nowadays, universities use technologies for a wide range of activities, such as communication with stakeholders, data collection and feedback. Using a process approach as a management strategy gives many advantages for the university. Defined, measured and improved processes lead to institutional development. Besides that, the university should measure the impact of processes on quality, safety and environmental factors. To do that, integration of systems is both a possibility and a must. Analysis of stakeholder groups and sufficient data collection is the key to systematic identification of needs. The identification of stakeholder needs contributes to understanding of what the university is offering to different stakeholder groups and what it must offer to assure future sustainability.

Developing the Riga Technical University excellence approach: case study

This research has focus on the development of the Riga Technical University (RTU) quality system – called "RTU excellence approach". Two main research studies were done before developing and introducing the RTU excellence approach as the first step of creating a sustainable quality system for the university (Table III).

In 2014, an empirical study of the RTU organisational culture was carried out. The organisational culture study was conducted in the context of continuous improvement regarding the university's development strategy and quality management. As a result, the first suggestions for creation and implementation of a quality system were introduced. The study results showed that RTU employees were ready for culture change, changes in processes and the development of a quality management system at the university (Lapiņa *et al.*, 2015).

In 2016, an assessment of RTU using the EFQM excellence model was performed. The self-assessment was made to detect the areas for improvement and to develop proposals for the RTU quality system formalisation using the model's Enabler criteria. A group of 12 experts was chosen from the main departments and levels of the university. As this study was focussed only on the model's Enabler criteria, the main conclusions were linked to what the university has done and what it is continuously doing to implement the RTU Strategy 2014-2020. The self-assessment results showed the three main priorities for RTU – focus on employees, effective process and partnership management (Medne and Lapiņa, 2018).

In 2017, the RTU excellence approach based on the principles of the EFQM excellence model and ESG (Standards and Guidelines for Quality Assurance in the European Higher Education Area, 2015) was developed (Iljins *et al.*, 2017). The RTU excellence approach reflects on the TQM basic principles, such as focus on stakeholders, system integration, process-orientation and continuous improvement.

In 2018, there was a research study using the full EFQM excellence model self-assessment methodology. The study included self-assessment, stakeholder analysis and prioritisation of the proposals. As a result, the main priorities for RTU were defined – the main performance analysis, employee and process management. After the determination of the main priorities, the University has implemented the given proposals in activity plans that are in line with the development of the new RTU strategy and the creation of a sustainable quality system.

Those three research studies helped to define the main RTU priorities for pursuing excellence. Each of the studies determined important aspects for the university's sustainable development and continuous improvement. The results of analysing the *Enabler* criteria of

Year	2014	2016	2017	2018	2019-2020
Activities	*An empirical study of the RTU organisational culture*	*The first assessment of RTU using the EFQM excellence model (Enabler criteria)*	*The development of the RTU excellence approach*	*The second self-assessment using the EFQM excellence model*	*The development of the RTU Strategy 2020 - 2025*
Results and practical implementation	Identified the main aspects of the RTU organisational culture	Identified the main problems and necessary improvements from the pilot project	Developed the stages of RTU excellence approach and the main priorities for sustainable development	Identified groups of stakeholders Identified the main problems and necessary improvements Compared the results with the first evaluation Prioritised proposals	Identified persons responsible for proposal implementation *Creating activity and development plans *Implementing proposals
Resulting priorities	(1) Culture change (2) Changes in processes (3) Development of a quality management system	(1) Focus on employees (2) Effective process management (3) Partnership management	Priorities are based on the EFQM excellence framework	(1) The main performance analysis (2) Employee management (3) Process management	Priorities are based on the EFQM excellence framework and self-assessments

Note: *Activities for 2019-2020

Table III. Main activities for developing sustainable quality system for RTU (created by authors)

the EFQM self-assessments showed that the focus should be on the following “3 P” priorities – partnerships, people and processes.

These priorities have been assessed to understand what are the main development focus areas from the perspective of RTU employees. This means that RTU employees have assessed partnerships, people and processes as the most important priorities for the uUniversity on the road to excellence. Dahlgaard-Park and Dahlgaard developed and introduced the “4 P” excellence framework model to pursue excellence that consists of five components – leadership, people, partnership, processes and products. With strong leadership, building excellence through people, partnerships and processes results in excellent products and services (Dahlgaard-Park, 2009). This means that RTU employees have determined similar components as priorities in the self-assessment. These priorities are taken into account while developing the new university’s Strategy. It indicates that the university has been able to build a solid foundation for development on the road to excellence.

The developed RTU excellence approach is a suitable solution for previously identified priorities. The framework is comprehensive and covers all of the priority areas of development – employee, process and stakeholder management. It has allowed a partial comparison of the university's progress within one year. In addition, employees involved in both self-assessments are now more familiar with the model. The organisational culture study at RTU revealed that employees are ready for change and the EFQM excellence model encourages changing and improving continuously. The planned self-assessments, proposal prioritising and implementation make it possible to comprehensively monitor the RTU processes and improve the main performance analysis. These are the main reasons why the EFQM excellence model is a good basis for a sustainable quality system and a long-term Strategy at RTU.

The excellence approach adoption has also resulted in the university's success at national and international rankings that assess a variety of sustainability elements. In the national level, RTU participates in a "sustainability index", which is organised by the Latvian Corporate Sustainability and Responsibility Institute. This assessment analyses the level of organisational sustainability and CSR. In 2018 and 2019, RTU was included in the highest – platinum – category.

At the international level, RTU participates in UI GreenMetric ranking that measures the university's performance in six categories. This ranking does not only analyse the environmental aspects, but also the overall management of the university. In 2018, RTU was ranked 128th in the world, which is one of the highest rankings for RTU. In 2019, RTU also participated in the Times Higher Education Impact Rankings and was ranked among the top 300 universities. This ranking is based on measuring the university's performance against the United Nations' sustainable development goals.

Sustainability elements through the focus of the Riga Technical University strategy

On the way to sustainable development, the previously mentioned sustainability elements must be integrated into the university`s strategy. Here, we discuss the strategic priorities set by RTU as an example. The next RTU strategy is being developed with a focus through the RTU excellence approach framework. The RTU excellence approach provides insight into the university's performance indicators, which can be compared with the performance indicators of the EFQM excellence model. The university's sustainable development is based on the RTU excellence approach. The RTU strategy, constitution and quality assurance policy are integrated with the RTU excellence approach. RTU has defined the main stages of the process leading to excellence by integrating Deming's PDCA/PDSA cycle (Figure 1).

The stages of RTU excellence approach reflect the RTU organisational culture and serve as a common language that ensures common understanding of the university quality issues, which in its turn, promotes sustainable development and achievement of the university goals (Riga Technical University, 2017).

The purpose of the analysis is to determine how the elements of sustainability are reflected in the priorities defined by the university and the university priorities define three key directions of RTU – study process, Research process and Innovation process with six horizontal tasks to be incorporated and implemented throughout the whole activity process of the university (Riga Technical University, 2014). The main priorities are as follows:

(1) Internationalisation: competitive activities of the university in the field of science, innovation and studies at international level.

(2) Interdisciplinarity: cooperation between different sectors and specialities, as a basis for development of new and innovative products and modern education content.

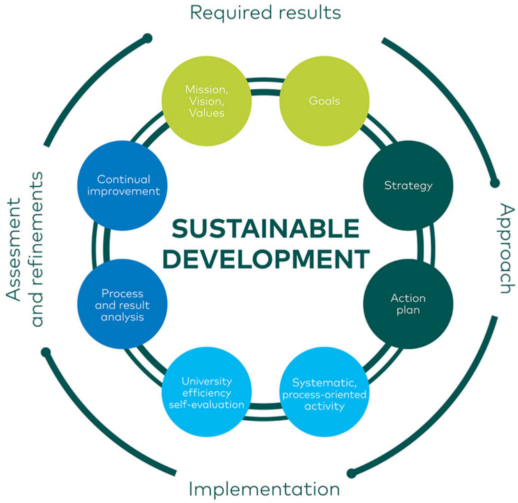

Figure 1.
Stages of RTU excellence approach
Riga Technical University (2017)

(3) Organisational efficiency: efficient and high quality management of the university to ensure development and modern implementation of the study and research processes.

(4) Financial efficiency: established financial independence and motivating internal financial system of the university to boost its growth.

(5) Infrastructure efficiency: up-to-date study, scientific and innovation environment with modern buildings and technical equipment that comply with the activities of the university.

(6) Smart digitalisation: The use of modern technologies to increase the efficiency of university activities (Riga Technical University, 2019).

Our experience already shows that the EFQM excellence model can be successfully integrated into the university's activities, providing an opportunity for targeted and regular self-assessment, identification of the necessary improvements and promotion of the development of the university's quality system, as well as overall improvement of the university's efficiency. The current strategic direction, core business processes and staff involvement in RTU contribute to the university's sustainable development. According to the RTU strategy and the main operational objectives, continuous improvement, implementation and monitoring of improvements are promoted. The following table shows the sustainability element overview through the focus of the university's strategy (Table IV).

This table is a guide to help higher education focus on sustainability through defined priorities in the university's strategy. It is important to have an understanding of sustainability as a wide concept at an international level. If this knowledge is integrated into the university's long-term strategy and it is part of organisational culture, then it is possible to engage internal stakeholders in different activities. Quality management and sustainability elements implemented in the organisation's strategy strengthen relationships with stakeholders, maintain sustainable project results and improve organisational culture (Frolova and Lapina, 2015).

Table IV. Sustainability elements through the focus of the RTU strategy (created by authors)

EFQM excellence model criteria/RTU priorities	1. Leadership	2. Strategy	3. People	4. Partnerships and resources	5. Processes, products and services
Internationalisation	Understanding of sustainability globally	Long term strategy objectives that are focussed on internationalisation	Staff and student engagement in international projects	Focus on sustainable stakeholders internationally	Benchmarking the best practices internationally
Interdisciplinarity	Understanding of sustainability in the context of modern education content	Focus on interdisciplinary projects	Staff and student engagement in interdisciplinary projects	Promoting collaboration with a wide range of partnerships	Providing modern education content related to sustainability
Organisational efficiency	Making sustainable decisions and defining objectives for sustainability	Integrated quality system and future oriented processes	Ensuring individual development plans for personnel	Use of integrated data system	Measuring and analysing the process impact on quality and sustainability
Financial efficiency	Sustainability aspects in finance planning	Consideration of finance aspects by developing a long term strategy	Ensuring clear personnel policy that promotes team work	Setting requirements for suppliers	Measuring the process impact on finance
Infrastructure efficiency	Understanding sustainability in infrastructure development	Consideration of environment in the infrastructure development plan	Ensuring safety and risk management activities in campus	Ensuring risk prevention in campus	Ensuring people-friendly infrastructure in campus
Smart digitalisation	Understanding of sustainability in the context of digitalisation	Focus on digitalisation in study, research and valorisation processes	Ensuring efficient and simple digital processes for employees; Providing additional training	Developing fully digital document management system	Ensuring digital process optimisation

The key aspect in developing sustainable management is benchmarking not only the best practices but also different activities related to sustainability. Interdisciplinarity related to the EFQM excellence model shows the understanding of sustainability in the context of providing modern education content. Developing study programmes with modern content creates value for students. Therefore, involving academic staff with knowledge about sustainability in the education content development process is a valuable asset for study programmes (Lapina *et al.*, 2016). Collaboration with a wide range of partners lets the university focus on interdisciplinary projects. Organisational efficiency includes development of an integrated quality system, reliable data system, measured and analysed processes – all with the purpose of organisational development and process optimisation. Financial efficiency as a strategic priority for the university consists of a set of performance indicators that are integrated into the strategy. These indicators need to show the overview of the university's main activities and how efficient they are. Infrastructure efficiency includes the understanding of sustainability and consideration of the environment; it is essential that infrastructure is people friendly, safe and sustainable.

Conclusions and discussions

Ensuring sustainability is a broad concept that has become an essential part of higher education. Research has shown that by analysing the university's environment, specifics and capabilities, it may find the best quality system development model for the institution. This paper analyses a variety of quality models and methods that are used in higher education institutions to improve their quality management. Although some authors are concerned about the business model implementation in the higher education context, the research reveals that universities have been able to integrate some elements of the business models in their management and it is possible to modify the EFQM excellence model to fit the needs of the higher education institution. One of the key successors is to integrate the EFQM excellence model approach and the university's strategy (Iljins *et al.*, 2017).

The results of the research confirm that a suitable quality management framework, such as the EFQM excellence model, gives more opportunities to develop the quality management system according to the needs of the university. Besides that, the EFQM excellence model recognises the main sustainability elements, mainly in the first five Enabler criteria. It is possible to integrate these sustainability elements in the strategy of higher education institution. The RTU strategy is given as an example for such integration.

By examining sustainability elements through the priorities defined in the strategy, it is possible to identify the activities to be carried out for sustainable development of the organisation. To provide an example, a comparison of RTU's strategy priorities with the *Enabler criteria* of the EFQM excellence model was performed. As a result, the activities to be carried out on the road to excellence were drawn up.

Answering the research question, we found that the sustainability of the university's quality system might be achieved by implementing the principles of the EFQM excellence model. In a way, universities can measure their performance with various international rankings that nowadays are introducing sustainability elements. Further discussion is needed to determine if the EFQM excellence model can provide universities with common performance indicators that are directly related to the University's sustainability or these indicators should be individual for each university and less likely to be benchmarked.

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